

Where the MST spectrum breaks: a cell-cloud theorem for token clouds and what it says about PHD-based AI-text detection

Companion note to *Multi-scale intrinsic dimension estimators for text embeddings*

MultiscalePHD

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Abstract

The persistent-homology dimension (PHD) of a token-embedding cloud is the growth exponent of the energy of its minimum spanning tree (MST). We prove that for a *cell cloud* — N compact cells of n points placed on a higher-dimensional frame — the sorted MST edge spectrum splits *exactly* at the quantile $q^* = (n - N)/(n - 1)$: below it lie only within-cell edges, above it only frame edges. The split point is a pure counting quantity; it does not depend on the geometry at all. If a cell is a token type, $q^* = 1 - \text{TTR}$ up to $O(1/n)$. We check this on HAP-E-2 (11 language models, 4 registers, ModernBERT-base embeddings): the observed break of the spectrum follows q^* with slope 0.99 ($r = 0.95$). The theorem then explains an old puzzle. Classical PHD, which sums the whole spectrum, tracks lexical diversity (rank correlation 0.94 with the AUROC of bigram diversity across models; sign agreement in 11/11 models), so its direction flips between lexically poor instruct models and lexically rich API models. Cutting the spectrum above q^* removes this: the band $[q_m^*, 1]$ with $q_m^* = q^* + \frac{1}{2}(1 - q^*)$ is almost independent of lexical statistics ($R^2 = 0.07$) and gives the same direction, AI above human, for all 11 models. Theorems are stated in the main text; proofs and further results (what each band estimates, the pole of the exponent-to-dimension map, validity tests) are in the appendices.

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1 The story in one page

The classical PHD estimator takes a point cloud, computes its MST, and reads the intrinsic dimension off the growth rate of the total edge length with the number of points. Its multiscale version keeps only edges whose lengths fall between two quantiles of the sorted spectrum and repeats the fit. The paper proves (its Theorem 1) that for a homogeneous cloud every such band gives the same exponent. That theorem is a statement of *harmlessness*: trimming does not spoil a consistent estimator. It also says that on a homogeneous cloud there is *no* multiscale signal — the profile “dimension versus scale” is flat — so whatever the bands show in practice measures departure from homogeneity. The question is what.

This note answers it for one specific, deliberately simple departure: a *cell cloud*. Points come in N compact cells of radius r , and the cell centres form a d_g -dimensional frame. Two things happen.

1. **The spectrum splits exactly (Theorem 3.1).** If cells are separated by more than $4r$, the MST is the union of the cells’ own MSTs and a frame tree on the cells, and the sorted spectrum breaks at rank $n - N$: the quantile

$$q^* = \frac{n - N}{n - 1}$$

separates within-cell edges (below) from frame edges (above). No geometric quantity enters: only the counts n and N .

2. **The two halves estimate different things (Appendix B).** A band inside the frame stratum estimates a β -diluted version of d_g , where β is the growth exponent of the number of cells with the number of points. A band inside the cell stratum, when cells are small, estimates no dimension at all: its value is a function of q^* and β only. A band straddling q^* does not interpolate, because the map exponent $\rightarrow$ dimension, $\kappa \mapsto \alpha/(1 - \kappa)$, has a pole at $\kappa = 1$ and the two strata lie on opposite sides of it.

For a token cloud the natural cell is a token type. Then N is the number of distinct types in the window, $N/n = \text{TTR}$ is the type–token ratio, and $q^* \approx 1 - \text{TTR}$. The counting quantities that drive the split are lexical statistics. This gives three predictions that need no new model runs: the break of the spectrum must sit at $1 - \text{TTR}$; any statistic that sums edges below q^* must track lexical diversity; a band that starts safely above q^* must not.

On HAP-E-2 all three hold. The observed break follows q^* with slope 0.99 (Fig. 1). Classical PHD is, for the purpose of human-versus-AI separation, a proxy of bigram diversity: its AUROC has rank correlation 0.94 with the AUROC of bigram diversity across the 11 models, and its *sign* flips accordingly — below 0.5 for lexically poor instruct models, above for lexically rich API models. The band $[q_m^*, 1]$, which starts halfway into the frame stratum, has $R_{\text{lex}}^2 = 0.07$ and $\text{AUROC} \geq 0.5$ for all 11 models. Its signal is moderate (AUROC 0.50–0.79) but its direction is stable, which is exactly what the theory predicts a frame-only statistic to be.

2 Setting

Throughout, $\|\cdot\|$ is the Euclidean norm in $\mathbb{R}^D$; D is the embedding width and never enters the answers.

2.1 MST, spectrum, trimmed energies, estimator

Definition 2.1 (MST and spectrum). For a finite set $X_n = \{x_1, \dots, x_n\} \subset \mathbb{R}^D$ with distinct pairwise distances, $\text{MST}(X_n)$ is the unique spanning tree of minimum total edge length; it has $n - 1$ edges. Its *spectrum* is the sorted sequence of edge lengths $\ell_{(1)} \leq \dots \leq \ell_{(n-1)}$. (The spectrum is the PH_0 barcode of the Vietoris–Rips filtration: Kruskal’s algorithm merges exactly the components that PH_0 merges.)

Definition 2.2 (Trimmed α -energy). For $\alpha > 0$ and $0 \leq p_1 < p_2 \leq 1$,

$$S_\alpha[p_1, p_2](X_n) = \sum_{i=\lfloor p_1(n-1) \rfloor + 1}^{\lfloor p_2(n-1) \rfloor} \ell_{(i)}^\alpha, \quad S_\alpha(X_n) = S_\alpha[0, 1](X_n) = \sum_{e \in \text{MST}(X_n)} |e|^\alpha.$$

$[p_1, p_2]$ is the *band*: a window of quantiles of the spectrum. α is a parameter of the method, not of the data; in practice $\alpha = 1$.

Definition 2.3 (Exponent and dimension estimate). For a family $\{X_n\}$ indexed by the number of points, the growth exponent of a band and the corresponding dimension estimate are

$$\kappa_{[p_1, p_2]} = \lim_{n \rightarrow \infty} \frac{\log S_\alpha[p_1, p_2](X_n)}{\log n}, \quad \hat{d}_{[p_1, p_2]} = \frac{\alpha}{1 - \kappa_{[p_1, p_2]}}.$$

Classical PHD is $\hat{d}_{[0, 1]}$. In practice the limit is replaced by a least-squares slope of $\log S_\alpha$ against $\log n'$ over a grid of subsample sizes n' drawn from one fixed cloud; we call the derivative $\kappa(n) = d \log S_\alpha / d \log n$ the *local exponent*.

Intuition. For n points spread over a d -dimensional set the typical nearest-neighbour distance is $n^{-1/d}$, so the MST length grows as $n \cdot n^{-1/d} = n^{1-1/d}$; the exponent $\kappa = 1 - 1/d$ is inverted to $d = 1/(1 - \kappa)$. The inversion has a pole: as $\kappa \rightarrow 1$ the estimate blows up, and for $\kappa > 1$ it is negative. This is not a numerical accident but a property of the formula, and it will matter (Theorem B.5).

The background result from the paper, restated: if X_n is an i.i.d. sample from a measure with density bounded away from 0 and ∞ on a compact subset of $\mathbb{R}^d$ and $0 < \alpha < d$, then $S_\alpha[p_1, p_2](X_n) = \Theta(n^{1-\alpha/d})$ for every fixed band, hence $\hat{d}_{[p_1, p_2]} = d$ for all bands simultaneously. We use it only through its Θ -form, and in Appendix B we extend that form to Ahlfors-regular measures (Lemma B.2), which is what the frame and the cells of a cell cloud are.

Definition 2.4 (Ahlfors regularity). A Borel probability measure μ on a compact $\mathcal{M} \subset \mathbb{R}^D$ is *d-regular* if $c_1 \rho^d \leq \mu(B(x, \rho)) \leq c_2 \rho^d$ for all $x \in \mathcal{M}$ and $0 < \rho \leq \rho_0$. (Uniform measures on compact d -manifolds and many fractals qualify.)

2.2 Cell cloud

Definition 2.5 (Cell cloud). A *cell cloud* $\mathcal{C}(N, \mathbf{k}, r; d_\ell, d_g)$ is built as follows.

- (a) *Frame.* Centres $C_1, \dots, C_N$ are an i.i.d. sample from a d_g -regular measure μ_g on a compact $\mathcal{M}_g \subset \mathbb{R}^D$ of unit diameter.
- (b) *Multiplicities.* Integers $k_1, \dots, k_N \geq 1$ with $\sum_i k_i = n$.
- (c) *Cells.* For each i , $Z_1^{(i)}, \dots, Z_{k_i}^{(i)}$ is an i.i.d. sample from a d_ℓ -regular measure μ_ℓ supported in the unit ball, and $X^{(i)} = \{C_i + rZ_j^{(i)}\}_{j \leq k_i}$.
- (d) *Cloud.* $X_n = \bigsqcup_i X^{(i)}$, $|X_n| = n$.

Definition 2.6 (Gap). Let $\delta = \min_{i \neq j} \|C_i - C_j\|$. The *gap* is $g = \delta/(2r)$: the smallest centre-to-centre distance in units of two cell radii. $g > 1$ means the cells do not overlap.

Assumption 2.7 (Separation). $g > 2$, i.e. $\delta > 4r$.

Intuition. Picture N small balls of radius r scattered on a d_g -dimensional surface, each holding k_i points spread in d_ℓ directions. Up close the cloud looks d_ℓ -dimensional, from afar d_g -dimensional, and between the balls it is empty — which is precisely why the paper’s homogeneity assumption fails. The factor 2 in the separation (rather than 1, no overlap) is a worst-case margin: it guarantees that the *longest* possible within-cell distance, $2r$, is shorter than the *shortest* possible between-cell distance, $\delta - 2r$. In practice the split appears well before $g = 2$.

3 Main result: the spectrum splits at a counting quantity

Theorem 3.1 (Exact splitting). *Let X_n be a cell cloud satisfying Assumption 2.7, with distinct pairwise distances. Then:*

- (i) $\text{MST}(X_n) = \left(\bigsqcup_{i=1}^N \text{MST}(X^{(i)}) \right) \sqcup E_{\text{fr}}$, where the frame E_{fr} has exactly $N - 1$ edges;
- (ii) as a graph on the cell indices, E_{fr} is the MST of the complete graph with single-linkage weights $w_{ij} = \text{dist}(X^{(i)}, X^{(j)})$;
- (iii) every within-cell edge has length $\leq 2r$ and every frame edge length $> 2r$;
- (iv) the spectrum splits at rank $n - N$: $\ell_{(i)} \leq 2r$ for $i \leq n - N$ and $\ell_{(i)} > 2r$ for $i > n - N$; hence the crossover quantile

$$q^* = \frac{n - N}{n - 1} = 1 - \frac{N - 1}{n - 1} = 1 - \frac{N}{n} + O\left(\frac{1}{n}\right)$$

is an exact identity, not an asymptotic one;

- (v) (band isolation) for any $0 \leq v_1 < v_2 \leq 1$ and $0 \leq u_1 < u_2 \leq 1$,

$$S_\alpha[v_1 q^*, v_2 q^*](X_n) = \text{trimmed energy of the forest } \bigsqcup_i \text{MST}(X^{(i)}) \text{ on its own band } [v_1, v_2],$$

$$S_\alpha[q^* + u_1(1 - q^*), q^* + u_2(1 - q^*)](X_n) = \text{trimmed energy of } E_{\text{fr}} \text{ on its own band } [u_1, u_2],$$

$$\text{in particular } S_\alpha[0, q^*] = \sum_i S_\alpha(X^{(i)}) \text{ and } S_\alpha[q^*, 1] = \sum_{e \in E_{\text{fr}}} |e|^\alpha.$$

Intuition. Kruskal’s algorithm adds edges in order of length. Because every within-cell pair is shorter than every between-cell pair, it first finishes all the cells — running, in effect, Kruskal inside each cell — and only then starts connecting cells, at which point the cheapest connection between two cells is their single-linkage distance. So the tree is “cells, then frame”, and in the sorted spectrum the first $n - N$ ranks are within-cell edges. The position of the break therefore depends on nothing but how many points and how many cells there are. Part (v) says that if bands are measured in coordinates relative to q^* (v inside the cell stratum, u inside the frame stratum), each band sits entirely in one stratum. The proof is in Appendix A.

Two companions make the theorem usable.

Proposition 3.2 (The frame is the tree on the centres, up to a constant). *Under Assumption 2.7, $\frac{1}{2}\|C_i - C_j\| \leq w_{ij} \leq \frac{3}{2}\|C_i - C_j\|$ for all $i \neq j$, and consequently*

$$2^{-\alpha} S_\alpha(\text{MST}(C_1, \dots, C_N)) \leq S_\alpha(E_{\text{fr}}) \leq (3/2)^\alpha S_\alpha(\text{MST}(C_1, \dots, C_N)),$$

and the same holds rank by rank for the sorted edge lengths of the two trees, hence for every band. A constant factor does not change a growth exponent, so for exponents the frame may be treated as the MST of the N centres.

Proposition 3.3 (q^* under subsampling). *Fix a cell cloud with multiplicities $k_1, \dots, k_N$ and draw a uniformly random subsample of size n' without replacement; let N' be the number of cells it hits and $q^*(n') = (n' - N') / (n' - 1)$. Then $m(n') = \mathbb{E}N'$ is concave in n' with $m(0) = 0$, $m(1) = 1$, and therefore*

$$n' \mapsto \bar{q}^*(n') := 1 - \frac{m(n') - 1}{n' - 1} \quad \text{is non-decreasing.}$$

In words: subsampling can only lower the crossover. A band $[q, 1]$ with $q \geq q^(n)$ computed once on the full cloud lies, in expectation, inside the frame stratum at every subsample size.*

Intuition. Under subsampling almost every surviving point ends up alone in its cell, so the type-token ratio of the subsample rises and the cell stratum shrinks toward zero. A cut placed above the full-cloud q^* therefore never re-enters the cell stratum on the way down the grid. Realised N' fluctuates around its mean and real clouds have a blurred transition (Section 5), which is why in practice the cut is placed with a margin: $q_m^* = q^* + \frac{1}{2}(1 - q^*)$, the midpoint of the frame stratum.

3.1 What the two strata estimate (statements; proofs in Appendix B)

The following results describe what a band returns once it is known which stratum it lies in. They assume a scaling regime (Assumption B.1): the number of cells grows as $N \asymp n^\beta$ with $\beta \in [0, 1]$ (for text, Heaps' law, $\beta \approx 0.6\text{--}0.8$), the cell radius as $r_n \asymp n^{-\gamma}$, and $0 < \alpha < \min(d_\ell, d_g)$.

- **Frame band is β -diluted (Theorem B.3).** Any band inside the frame stratum has $\kappa_{\text{fr}} = \beta(1 - \alpha/d_g)$ and $\hat{d}_{\text{fr}} = \alpha d_g / (d_g(1 - \beta) + \alpha\beta)$, which equals d_g iff $\beta = 1$ and is strictly smaller otherwise. Reason: the frame lives on N points but the estimator divides by $\log n$. The cure is to regress on $\log N'$ instead (Corollary B.11).
- **Cell band with small cells is not a dimension (Theorem B.4).** If cells hold at most K points and r does not shrink, the local exponent of $[0, q^*]$ is $\kappa_{\text{cell}} = (1 - \beta_n(1 - q^*)) / q^* > 1$, so $\hat{d}_{\text{cell}} = -\alpha q^* / ((1 - \beta_n)(1 - q^*)) < 0$: a function of counting quantities only, with no geometry in it.
- **A mixed band does not interpolate (Theorem B.5).** When the two strata have exponents on opposite sides of 1, a band containing both returns a value *outside* the interval between the two stratum estimates — possibly negative, possibly huge — because $\kappa \mapsto \alpha / (1 - \kappa)$ passes through a pole in between. A fixed band such as $[0, 0.5]$ or $[0.5, 1]$ switches stratum whenever q^* crosses its boundary (Corollary B.7).
- **Two free validity tests (Corollaries B.8 and B.9).** A true dimension does not depend on α ; $\hat{d}_{\text{fr}}$ does unless $\beta = 1$. A true dimension $d > \alpha$ gives $\hat{d} > \alpha$; any band with $\hat{d} \leq \alpha$ estimates no dimension.
- **Corrected estimator (Proposition B.10).** (i) Recompute q^* on *each* subsample and place bands relative to it; (ii) fit the frame band against $\log N'$. On synthetic cell clouds this pair reduces the response to the number of cells at frozen geometry from $\times 8.1$ (fixed band $[0.5, 1]$) to $\times 1.3$, and raises the elasticity with respect to d_g from 0.10 to 0.89 (Appendix C).

4 From cells to text

Everything above is about a geometric object we defined ourselves. To use it on token clouds four hypotheses are needed; each is falsifiable and each was tested in Section 5.

Hypothesis 1 (Token clouds are cell clouds). Contextual token embeddings of one document form compact, non-overlapping groups whose centres lie on a higher-dimensional frame.

Hypothesis 2 (A cell is a token type). All occurrences of one token type lie in one cell. Then N is the number of distinct types in the window, $\text{TTR} = N/n$, and by Theorem 3.1(iv)

$$q^* = 1 - \text{TTR} + O(1/n).$$

Hypothesis 3 (Heaps’ law). The number of types grows as $N \asymp n^\beta$ with $\beta < 1$; on a subsampling grid the local β_n is close to 1 at small n' and smaller at large n' .

Hypothesis 4 (Sparse regime). In a window of a few hundred tokens the mean multiplicity $\bar{k} = 1/\text{TTR} \approx 1.5\text{--}2$ is bounded, so the cell stratum is in the regime of Theorem B.4, and cells do not shrink with window length ($\gamma = 0$).

Intuition. If these hold, the entire lexical content of a PHD-type statistic sits in two counting quantities, $q^* = 1 - \text{TTR}$ and β . Three consequences are testable on data that already exist. (P1) The break of the spectrum, found from the token labels, must coincide with $1 - \text{TTR}$ computed from counts alone. (P2) Any statistic that sums edges below q^* — classical PHD included, since its spectrum starts at 0 — must co-vary with lexical diversity, and by Theorem B.5 its sign relative to a reference can go either way. (P3) A band that starts above q^* on every subsample (Proposition 3.3) must be free of that dependence. The direction of a comparison between two *different* texts (human versus model) is a class-B comparison in the sense of Appendix B: counting quantities differ between the groups, so any signal must be checked against lexical statistics; the HAP-E-2 design does exactly this.

5 HAP-E-2: PHD and lexis, split at q^*

5.1 Data

Corpus. HAP-E-2 (Human–AI Parallel English Corpus 2; Reinhart et al., PNAS 2025, repository version 2026). Each source document is cut into two halves of about 500 words; models receive the first half and write a continuation, and the human continuation is the second half of the same document. Models: Llama-3 8B/70B and Gemma-2 9B/27B (base and instruct), GPT-4o, GPT-4o-mini, GPT-5-mini; Claude Haiku 4.5 is excluded (below).

Registers. Register = text type, following COCA. Used: acad (open-access Elsevier articles), news (US online news), blog (blogger.com), fic (Project Gutenberg fiction). Not used: spok (podcast transcripts) and tvn (screenplays, partly OCR).

Sample. In each register 300 documents were drawn at random among those where every author has at least 128 tokens: 1200 human texts and 13200 model texts.

Claude Haiku 4.5. In the published files its texts do not match their doc_id: the TF-IDF-nearest human text is the “own” document in 0.2% of cases, against 51.2% for GPT-4o and 46.5% for Llama-3-8B-Instruct (mean similarity to the own document 0.02 vs 0.21 and 0.25). Paired comparisons are impossible, so the model is dropped.

Preprocessing. Identical for all authors: typographic quotes and dashes mapped to ASCII, COCA-style spaces before punctuation removed (~ 1.3 per 100 words in human news, near zero for API models), markdown markers stripped. The first 128 tokens are taken; each window is encoded separately with ModernBERT-base, last hidden layer. Attention-sink tokens (2.6%: *the*, full stop, comma with norm ≈ 26) are merged into one pseudo-type.

5.2 Methods

Spectrum and q^* . For each window the exact Euclidean MST is built and its $n - 1$ edges sorted. The type of a token is its id. By Theorem 3.1,

$$q^* = \frac{n - N}{n - 1} \approx 1 - \text{TTR}, \quad q_m^* = q^* + \frac{1}{2}(1 - q^*),$$

where q_m^* is q^* “with a margin”: the midpoint of the frame stratum. On HAP-E-2, $q^* \in [0.15, 0.45]$ (1st–99th percentile), so $q_m^* \in [0.57, 0.72]$.

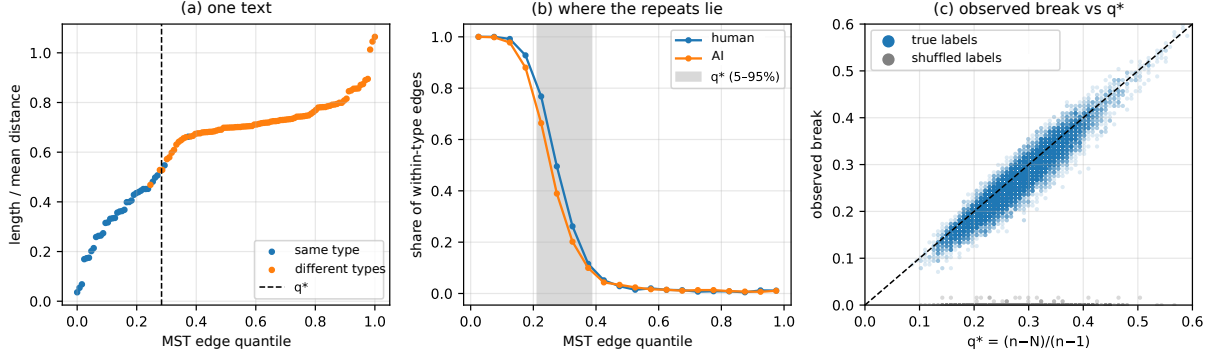


Figure 1: Cell structure of the ModernBERT cloud. (a) Sorted MST edges of one human text: edges between two occurrences of the same type (blue) occupy the short end of the spectrum up to q^* . (b) Share of within-type edges by spectrum quantile, averaged over 400 human and 400 AI texts; grey band: 5–95% range of q^* . (c) Observed break (the quantile that best separates within-type from between-type edges) against q^* for all 14400 texts; grey: the same clouds with type labels shuffled within the text.

Band PHD. For a band $[a, b]$, $S_{[a,b]}(n')$ is the sum of the MST edge lengths of a subsample of n' tokens that fall between quantiles a and b of the subsample’s sorted spectrum ($\alpha = 1$). Subsample sizes $n' \in \{21, 41, 61, 81, 101, 121\}$, 32 subsamples per size without replacement; κ is the slope of $\log \bar{S}$ against $\log n'$ and $d = 1/(1 - \kappa)$. Bands: $[0, 1]$ (classical PHD), $[0, q_m^*]$ and $[q_m^*, 1]$; q^* and q_m^* are computed on the full window and kept fixed for all its subsamples, which by Proposition 3.3 keeps $[q_m^*, 1]$ inside the frame stratum at every n' . Variants with q^* recomputed per subsample and with the $\log N'$ fit are reported in Table 1.

Lexical measures. TTR, hapax share (tokens whose type occurs once in the window) and bigram diversity (share of distinct adjacent-token pairs); the last one sees repeated short phrases that TTR does not.

Evaluation. AUROC = $P(\text{statistic of AI} > \text{statistic of human})$ for each model against the human texts of the same registers; 0.5 means no direction. 95% confidence intervals: bootstrap over documents within registers (1000 replicates). Lexical dependence: R^2 of a regression of the statistic on TTR and hapax share over human texts. Lexically matched pairs: human/model pairs of the same document whose TTR, hapax share and bigram diversity differ by at most 0.02 (630 pairs).

5.3 The token cloud is a cell cloud (P1)

If a cell is a token type, the break of the spectrum must coincide with q^* computed from counts alone. It does (Fig. 1c): slope 0.99, $r = 0.95$, mean offset -0.012. The check is not trivial: with shuffled labels the slope is 0.00 ($r = 0.03$). Separation is not perfect — the median leakage is 8% (share of the $n - N$ shortest edges that join different types), and the transition from repeats to frame is blurred over quantiles ≈ 0.2 – 0.45 (Fig. 1b). Above 0.5 there are essentially no within-type edges. This blur is the empirical reason for the margin in q_m^* .

5.4 Classical PHD tracks lexis (P2)

The sign of the classical-PHD AUROC agrees with the sign of the bigram-diversity AUROC for 11 of 11 models (with TTR: 10) and in 43 of 44 model $\times$ register cells; the rank correlations are 0.94 over models and 0.97 over cells (Fig. 2, left). Instruct versions of Llama-3 and Gemma-2 write lexically poorer than humans — and their PHD is lower; API models and base Gemma write richer — and their PHD is higher. A linear model of PHD on TTR and hapax share, fitted on human texts only,

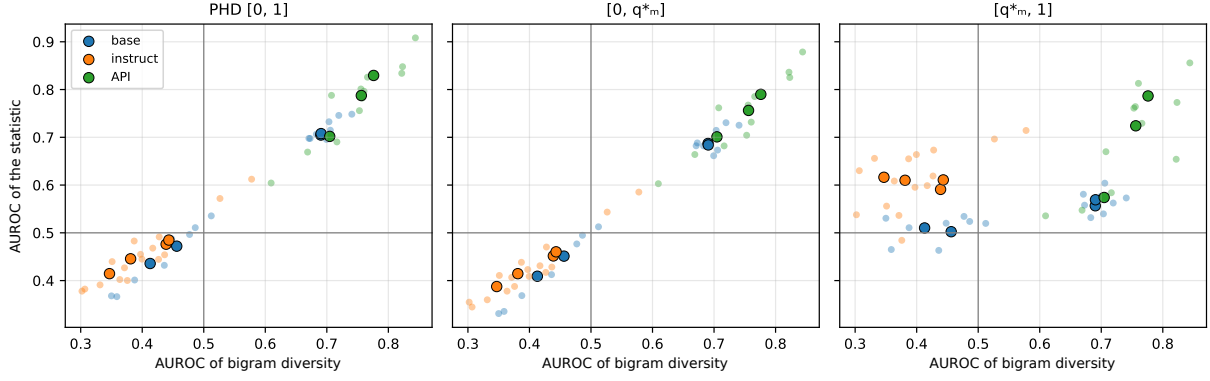


Figure 2: AUROC of the statistic against AUROC of bigram diversity. Large markers: models (4 registers pooled); small: model $\times$ register; colour: model type.

statistic	R_{lex}^2	ρ_S	> 0.5	base	instruct	API
PHD $[0, 1]$	0.61	0.94	5/11	0.58	0.46	0.77
$[0, q_m^*]$	0.61	0.95	5/11	0.56	0.43	0.75
$[q_m^*, 1]$	0.07	0.20	11/11	0.53	0.61	0.69
$[0, q^*]$ (no margin)	0.10	0.45	3/11	0.37	0.48	0.53
$[q^*, 1]$ (no margin)	0.17	0.75	9/11	0.51	0.56	0.73
$[0, 0.7]$ (fixed)	0.69	0.96	5/11	0.58	0.42	0.76
$[0.7, 1]$ (fixed)	0.07	0.19	11/11	0.54	0.60	0.69
$[q^{*'}, 1]$, slope vs $\log n'$	0.85	0.90	7/11	0.68	0.44	0.81
$[q^{*'}, 1]$, slope vs $\log N'$	0.05	0.17	8/11	0.48	0.63	0.70

Table 1: Band variants. R_{lex}^2 : share of the statistic’s variance over human texts explained by TTR and hapax share; ρ_S : Spearman correlation between the AUROC of the statistic and the AUROC of bigram diversity over the 11 models; > 0.5 : number of models with AUROC above 0.5; base / instruct / API: mean AUROC in each model group. $q^{*'}: q^*$ recomputed on every subsample.

predicts the AI–human difference across cells with $r = 0.89$; with bigram diversity, $r = 0.94$ at slope 1.02. In lexically matched pairs classical PHD does not separate human from model: $P = 0.52$.

5.5 Splitting the spectrum at q_m^* (P3)

The lower band $[0, q_m^*]$ behaves like classical PHD: $R_{\text{lex}}^2 = 0.61$, $\rho_S = 0.95$, only 5 models above 0.5 (Fig. 2, Table 1). The upper band $[q_m^*, 1]$ is almost free of lexis: $R_{\text{lex}}^2 = 0.07$, $\rho_S = 0.20$, and all 11 models have AUROC ≥ 0.5 .

The margin matters. A band starting exactly at q^* captures the blurred transition zone, and its size again depends on TTR: $[q^*, 1]$ gives $R_{\text{lex}}^2 = 0.17$, $\rho_S = 0.75$ and 9 models out of 11. The fixed band $[0.7, 1]$ is equivalent to $[q_m^*, 1]$ ($\rho_S = 0.19$, 11/11) because q_m^* lies in the narrow range 0.57–0.72 on this corpus. Recomputing q^* on every subsample and fitting against $\log n'$ turns the slope into the growth exponent of the number of types, as Theorem B.3 predicts ($R_{\text{lex}}^2 = 0.85$); the theoretical correction — the same band fitted against $\log N'$ — removes lexis again ($R_{\text{lex}}^2 = 0.05$) but pushes the base models below 0.5. For *detection* the margin band is the better tool; the corrected estimator is the tool for *absolute* dimension, which is not the question here.

Figure 3 shows both bands at once. By AUROC the models fall into two groups: for 6 models the lower band is below the human level and the upper band above (the 4 instruct models and the 2 base Llamas, whose upper band is near 0.5); for 5 models (API and base Gemma) both are above. In absolute values (Fig. 3b) instruct models sit up-left of the humans and API models up-right: the lower band follows the direction of lexis, the upper band is never below the human level (base Llamas

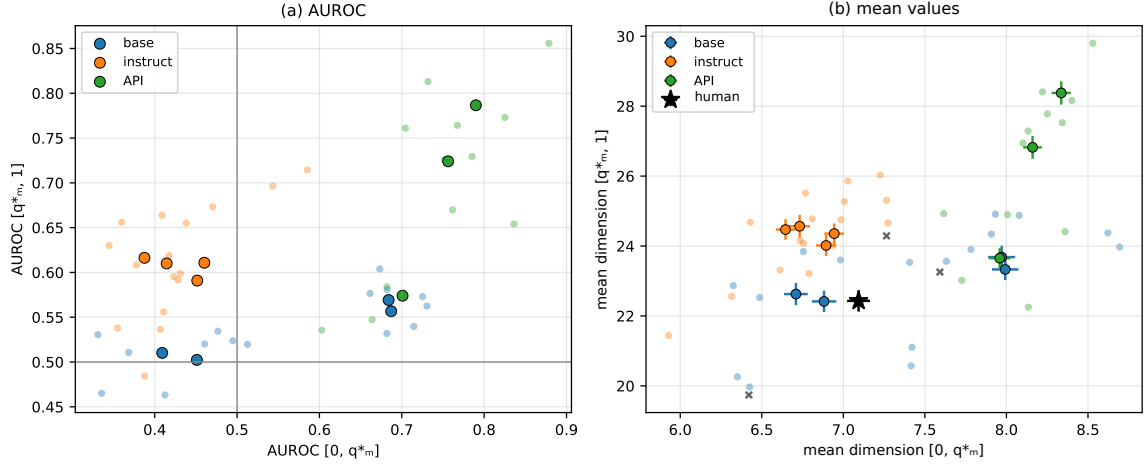


Figure 3: Lower band against upper band. (a) AUROC of the two bands; large markers: models, small: model $\times$ register. (b) Mean dimension values of the two bands; crosses: 95% intervals, star: humans, grey crosses: humans by register.

model	type	TTR	bigrams	PHD $[0, 1]$	$[0, q_m^*]$	$[q_m^*, 1]$	$[0.7, 1]$
Llama-3-8B	base	0.50	0.41	0.44	0.41	0.51	0.51
Llama-3-70B	base	0.55	0.46	0.47	0.45	0.50	0.51
Gemma-2-9B	base	0.85	0.69	0.71	0.68	0.57	0.57
Gemma-2-27B	base	0.84	0.69	0.70	0.69	0.56	0.56
Llama-3-8B-Instruct	instruct	0.29	0.35	0.41	0.39	0.62	0.61
Llama-3-70B-Instruct	instruct	0.34	0.38	0.45	0.41	0.61	0.61
Gemma-2-9B-it	instruct	0.40	0.44	0.48	0.46	0.61	0.61
Gemma-2-27B-it	instruct	0.41	0.44	0.48	0.45	0.59	0.59
GPT-4o	API	0.77	0.76	0.79	0.76	0.72	0.72
GPT-4o-mini	API	0.79	0.78	0.83	0.79	0.79	0.78
GPT-5-mini	API	0.74	0.70	0.70	0.70	0.57	0.57

Table 2: AUROC ($P(\text{AI} > \text{human})$) by model, 4 registers pooled. The columns TTR and bigrams are the AUROC of the lexical statistics themselves.

are at it).

5.6 Per model

In Fig. 4 classical PHD and the lower band $[0, q_m^*]$ lie on both sides of 0.5, depending on whether the model writes lexically poorer or richer than humans. The upper band $[q_m^*, 1]$ is at or above 0.5 for every model (minimum AUROC 0.50, lowest interval bound 0.48, both for base Llama-3). In lexically matched pairs the upper band keeps its direction: $P(\text{AI higher}) = 0.66$ for instruct models and 0.64 for API (± 0.07), 0.48 for base models; classical PHD gives 0.61, 0.55 and 0.39.

The price: where lexis and geometry act in the same direction (API models, base Gemma) the upper band separates less strongly than classical PHD (Table 2). Its advantage is a stable direction, not strength.

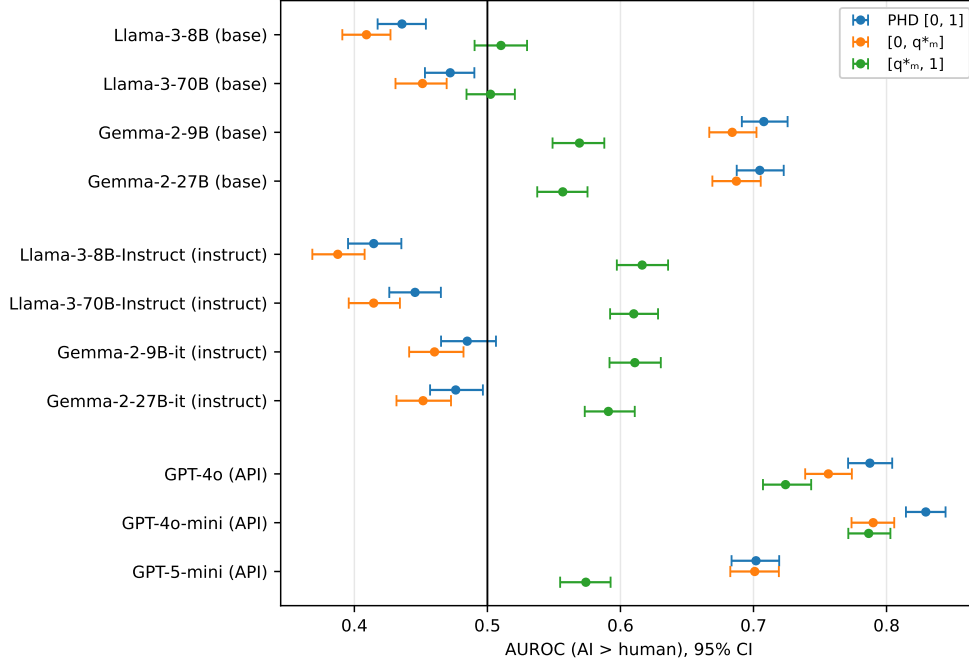


Figure 4: AUROC of each model with 95% confidence interval (bootstrap over documents). Vertical line: 0.5.

5.7 Conclusions

1. The last-layer ModernBERT cloud is a cell cloud with cell = token type: the break of the MST spectrum is predicted by counts alone, $q^* = (n - N) / (n - 1)$, as Theorem 3.1 requires.
2. Classical PHD, human versus AI, is essentially a measure of lexical diversity with phrase repeats included. Hence its sign depends on the generator: below human for lexically poor instruct models, above for rich API models. This is why PHD does not fix an order between human and AI texts — and by Theorem B.5 a statistic that mixes the two strata is not expected to.
3. Splitting the spectrum at q^* isolates the lexical part: it lives in $[0, q_m^*]$. The band $[q_m^*, 1]$, starting with a margin above q^* , is lexis-free and gives one direction (AI above) for all 11 models. The effect is moderate (AUROC 0.50–0.79) and stronger for fine-tuned than for base models.
4. In practice a fixed start at 0.7 suffices: on this corpus it is equivalent to the adaptive q_m^* because q_m^* varies little; on corpora with a wider TTR range the adaptive cut is the safe choice.

5.8 Limitations

- One embedding model (ModernBERT-base, last layer) and a 128-token window. The paper’s experiments use decoder states (Gemma-4-E4B-it); the cell structure has to be re-measured there before numbers are transferred.
- HAP-E-2 asks models to continue a text in the human’s style; decoding settings are not recorded in the corpus.
- Claude Haiku 4.5 is excluded because its documents are not paired.
- The residual signal of the upper band is partly stylistic: fine-tuned models use fewer names and numbers, which form dense groups in the cloud.
- The upper band with fixed q_m^* is a *drifting* frame band (its relative position moves as n' decreases), so its absolute value is not d_g (Theorem B.3); for the detection question only its direction is used.

6 What to do with this

1. **Publish q^* with every PHD-type number.** It costs a token count and tells the reader which stratum a band is in.
2. **For detection, use a band that starts above $q^* - [q_m^*, 1]$** or, on corpora with a narrow TTR range, a fixed start around 0.7. Report the lexical baseline (TTR, hapax share, bigram diversity) next to it; a statistic that does not beat the baseline is measuring lexis.
3. **For absolute dimension, use the corrected estimator** (Proposition B.10): q^* recomputed per subsample, frame band fitted against $\log N'$, and apply the sign and α tests. Expect the result to be limited by conditioning: at $d \approx 24$ a 1% systematic drift of the energy along the grid moves the estimate by a third (Appendix C).
4. **Treat experiments that change the text and experiments that change the representation differently.** When the text is fixed (layers, checkpoints, quantisation), n, N, q^* and β are identical by construction and the counting confound is zero; when the text changes (human vs AI, domain vs domain) it is not, and matching on TTR is the minimum.

A Proofs of the main results

Lemma A.1 (MST is optimal for every monotone criterion). *Let $T^* = \text{MST}(X_n)$ and T any spanning tree; sort the edge lengths of each, $t_{(1)}^* \leq \dots \leq t_{(n-1)}^*$ and $t_{(1)} \leq \dots \leq t_{(n-1)}$. Then $t_{(k)}^* \leq t_{(k)}$ for all k , and hence $\sum_{e \in T^*} f(|e|) \leq \sum_{e \in T} f(|e|)$ for every non-decreasing $f \geq 0$.*

Proof. Fix k and put $t = t_{(k)}$. T has k edges of length $\leq t$, so the subgraph $T \cap \{|e| \leq t\}$ has $n - k$ components; hence the threshold graph $G_t = \{(x, y) : \|x - y\| \leq t\}$ has at most $n - k$ components. Kruskal's algorithm, having scanned all edges of length $\leq t$, has connected everything G_t connects, i.e. has accepted at least k edges of length $\leq t$; so $t_{(k)}^* \leq t$. The sum inequality follows termwise. $\square$

Lemma A.2 (Separation gives a gap in the spectrum). *Under Assumption 2.7, $\max_i \text{diam}(X^{(i)}) \leq 2r < \delta - 2r \leq \min_{i \neq j} \text{dist}(X^{(i)}, X^{(j)})$.*

Proof. Points of $X^{(i)}$ lie in $\bar{B}(C_i, r)$, so $\text{diam } X^{(i)} \leq 2r$. For $x \in X^{(i)}, y \in X^{(j)}, i \neq j$: $\|x - y\| \geq \|C_i - C_j\| - \|x - C_i\| - \|y - C_j\| \geq \delta - 2r$, and $\delta > 4r \iff \delta - 2r > 2r$. $\square$

Proof of Theorem 3.1. Run Kruskal's algorithm on the complete graph on X_n : scan pairs in increasing order of distance, accept a pair iff it joins two different components. By Lemma A.2 every within-cell pair is scanned before every between-cell pair.

Phase 1. While within-cell pairs are scanned, no accepted edge joins two cells, so the components containing points of $X^{(i)}$ consist of points of $X^{(i)}$ only. The restriction of the run to pairs inside $X^{(i)}$ is exactly Kruskal on $X^{(i)}$, whose output is $\text{MST}(X^{(i)})$ with $k_i - 1$ edges. At the end of Phase 1 the components are precisely the cells.

Phase 2. Between-cell pairs are scanned; joining components now means joining cells. This is Kruskal on the multigraph obtained by contracting every cell to a vertex; since pairs come in increasing order, the first pair scanned between two given cells is the shortest one, of length $w_{ij} = \text{dist}(X^{(i)}, X^{(j)})$. Kruskal on the contracted graph is Kruskal on the complete graph with weights w_{ij} , which gives (ii); connecting N vertices takes exactly $N - 1$ edges. Edge count: $\sum_i (k_i - 1) + (N - 1) = n - 1$, which is (i).

(iii) is Lemma A.2. (iv): exactly $n - N$ edges are $\leq 2r$ and the rest are $> 2r$, so after sorting the first $n - N$ ranks are within-cell edges; dividing by $n - 1$ gives q^* . The expansion follows from $\frac{N-1}{n-1} - \frac{N}{n} = \frac{N-n}{n(n-1)} \in [-\frac{1}{n-1}, 0]$.

(v): by (iv), ranks $1, \dots, n - N$ are exactly the within-cell edges and ranks $n - N + 1, \dots, n - 1$ exactly the frame edges. The band $[v_1 q^*, v_2 q^*]$ in quantiles of the whole cloud is the rank range

$[v_1(n - N), v_2(n - N)]$, i.e. the relative band $[v_1, v_2]$ of the first block; likewise for the second block. The sums agree term by term. $\square$

Proof of Proposition 3.2. By the triangle inequality $D_{ij} - 2r \leq w_{ij} \leq D_{ij} + 2r$ with $D_{ij} = \|C_i - C_j\|$; under Assumption 2.7, $2r < \delta/2 \leq D_{ij}/2$, hence $\frac{1}{2}D_{ij} \leq w_{ij} \leq \frac{3}{2}D_{ij}$. Let $T_w = E_{\text{tr}}$ (MST for weights w) and $T_D = \text{MST}(C)$ (MST for weights D). By Lemma A.1 applied with weights w , the k -th smallest w -length of T_w is at most the k -th smallest w -length of T_D , which is at most $\frac{3}{2}$ times the k -th smallest D -length of T_D (a monotone map of a multiset preserves the order of its order statistics). Symmetrically, with weights D , the k -th smallest D -length of T_D is at most that of T_w , which is at most 2 times the k -th smallest w -length of T_w . So the sorted spectra of the two trees agree rank by rank within the factor $[\frac{1}{2}, \frac{3}{2}]$, and so do all band energies within $[2^{-\alpha}, (3/2)^\alpha]$. $\square$

Proof of Proposition 3.3. Write the subsample as the first n' positions of a uniformly random permutation of the n points, and let T_i be the position of the first occurrence of cell i . Then $N' = \#\{i : T_i \leq n'\}$ and $m(n') = \sum_i \Pr(T_i \leq n')$, so

$$m(n' + 1) - m(n') = \sum_i \Pr(T_i = n' + 1), \quad \Pr(T_i = j) = \binom{n-j}{k_i-1} / \binom{n}{k_i},$$

because the other $k_i - 1$ copies must occupy positions after j . This is non-increasing in j , so the increments of m are non-increasing: m is concave. Clearly $m(0) = 0$ and $m(1) = 1$. For a concave function the slope of the chord from the fixed point $(1, m(1))$ to $(n', m(n'))$, namely $(m(n') - 1)/(n' - 1)$, is non-increasing in n' ; hence $\bar{q}^*(n') = 1 - (m(n') - 1)/(n' - 1)$ is non-decreasing. Finally, a subsample of a cell cloud is a cell cloud with the same centres and radius, so Theorem 3.1 applies to it, and a cut at $q \geq q^*(n) \geq \bar{q}^*(n')$ lies above the subsample's expected crossover. $\square$

B Further results: what each band estimates

Assumption B.1 (Scaling regime). A sequence of cell clouds indexed by n with (a) $\log N(n)/\log n \rightarrow \beta \in [0, 1]$; (b) $-\log r_n/\log n \rightarrow \gamma \geq 0$; (c) balanced multiplicities, $c\bar{k} \leq k_i \leq C\bar{k}$, $\bar{k} = n/N$; (d) $\mu_g d_g$ -regular, $\mu_\ell d_\ell$ -regular; (e) Assumption 2.7 for all n ; (f) $0 < \alpha < \min(d_\ell, d_g)$.

Intuition. $\beta = 1$ means new points mostly open new cells; $\beta = 0$ means they pile into existing ones. $\gamma = 0$ means cells keep their size as the sample grows. The self-similar case $\beta = 1$, $\gamma = 1/d_g$, $d_\ell = d_g$ recovers a homogeneous cloud, and all formulas below collapse to $\hat{d} = d$ — the paper's Theorem 1 as a special case.

Lemma B.2 (Θ -version of trimming invariance for regular measures). *Let μ be d -regular on a compact $\mathcal{M} \subset \mathbb{R}^D$, X_n i.i.d. from μ , $0 < \alpha < d$, $0 \leq p_1 < p_2 \leq 1$ fixed. Then there are constants $0 < c \leq C$ with $c n^{1-\alpha/d} \leq S_\alpha[p_1, p_2](X_n) \leq C n^{1-\alpha/d}$ with probability tending to one. Hence $\kappa_{[p_1, p_2]} = 1 - \alpha/d$ in probability.*

Proof. Upper bound. $S_\alpha[p_1, p_2] \leq S_\alpha(X_n)$, and for any n points of the support of a d -regular measure $S_\alpha \leq C n^{1-\alpha/d}$ when $\alpha < d$ — a deterministic covering bound (the upper box dimension of the support is d), see Schweinhart [3].

Lower bound. Put $m = n - 1$, $p' = (p_1 + p_2)/2$. The band contains at least $\frac{1}{2}(p_2 - p_1)m - 1$ edges, each no shorter than $\ell_{(\lfloor p'm \rfloor)}$. It remains to show $\ell_{(\lfloor p'm \rfloor)} \geq s_n := c_0 n^{-1/d}$ with probability $\rightarrow 1$ for a suitable c_0 , i.e. that fewer than $p'm$ MST edges are shorter than s_n . Each such edge is a pair of points at distance $< s_n$ and distinct edges are distinct pairs, so their number is at most $U_n = \#\{\{i, j\} : \|X_i - X_j\| < s_n\}$. By regularity $\Pr(\|X_i - X_j\| < s_n) \leq c_2 s_n^d = c_2 c_0^d/n$, so $\mathbb{E}U_n \leq \frac{1}{2}c_2 c_0^d n$; choose c_0 with $\frac{1}{2}c_2 c_0^d \leq p'/4$, giving $\mathbb{E}U_n \leq p'm/2$. U_n is a U -statistic of order 2 with indicator kernel; its variance is $\binom{n}{2}[2(n-2)\zeta_1 + \zeta_2]$ with $\zeta_2 \leq \Pr(h=1) = O(1/n)$ and $\zeta_1 \leq \mathbb{E}[\mu(B(X_1, s_n))^2] = O(1/n^2)$, so $\text{Var } U_n = O(n)$ and by Chebyshev $\Pr(U_n \geq p'm) \leq O(n)/(p'm/2)^2 \rightarrow 0$. $\square$

Theorem B.3 (Frame band). *Under Assumption B.1, for any fixed $0 \leq u_1 < u_2 \leq 1$,*

$$S_\alpha[q^* + u_1(1 - q^*), q^* + u_2(1 - q^*)](X_n) = \Theta(N^{1-\alpha/d_g}),$$

$$\kappa_{\text{fr}} = \beta \left(1 - \frac{\alpha}{d_g}\right), \quad \hat{d}_{\text{fr}} = \frac{\alpha d_g}{d_g(1 - \beta) + \alpha \beta}.$$

$\beta \mapsto \hat{d}_{\text{fr}}$ increases strictly from α at $\beta = 0$ to d_g at $\beta = 1$; $\hat{d}_{\text{fr}} = d_g$ iff $\beta = 1$.

Proof. By Theorem 3.1(v) the band is the relative band $[u_1, u_2]$ of E_{fr} , and by Proposition 3.2 its energy is within a constant factor of the same relative band of $\text{MST}(C_1, \dots, C_N)$. The centres are N i.i.d. points from a d_g -regular measure, so Lemma B.2 with N in place of n gives $\Theta(N^{1-\alpha/d_g})$. Dividing logarithms by $\log n$ and using $\log N / \log n \rightarrow \beta$: $\kappa_{\text{fr}} = \beta(1 - \alpha/d_g)$. Then $\hat{d}_{\text{fr}} = \alpha / (1 - \beta(1 - \alpha/d_g)) = \alpha d_g / (d_g - \beta(d_g - \alpha))$; the denominator decreases in β since $d_g > \alpha$. $\square$

Theorem B.4 (Cell band). (a) Dense regime, $\bar{k} \rightarrow \infty$: under Assumption B.1,

$$S_\alpha[0, q^*](X_n) = \Theta(r_n^\alpha N^{\alpha/d_\ell} n^{1-\alpha/d_\ell}), \quad \kappa_{\text{cell}} = 1 - \alpha\gamma - (1 - \beta)\frac{\alpha}{d_\ell}, \quad \hat{d}_{\text{cell}} = \frac{1}{\gamma + (1 - \beta)/d_\ell},$$

independent of α , equal to d_ℓ iff $\gamma = \beta = 0$, and equal to $1/\gamma$ (no d_ℓ at all) when $\beta = 1$.

(b) Sparse regime, a local statement on a finite range of n under: (c1) $k_i \leq K$; (c2) $N(n)$ smooth with local exponent $\beta_n = d \log(N - 1) / d \log n$; (c3) the mean energy of one within-cell edge, $c_n = S_\alpha[0, q^*] / (r_n^\alpha (n - N))$, has negligible log-derivative ε_n . Then the local exponent is

$$\kappa_{\text{cell}}(n) = \frac{1 - \beta_n(1 - q^*)}{q^*} - \alpha\gamma + \varepsilon_n,$$

and for $\gamma = 0$, $\varepsilon_n = 0$, $\beta_n < 1$: $\kappa_{\text{cell}} > 1$ and $\hat{d}_{\text{cell}} = -\alpha q^* / ((1 - \beta_n)(1 - q^*)) < 0$. The premises (c1) and $\beta_n < 1$ are compatible only on a finite range: $k_i \leq K$ forces $N \geq n/K$, hence $q^* \leq 1 - 1/K$ and $\beta_n \rightarrow 1$ on average.

Proof. (a) By Theorem 3.1(v), $S_\alpha[0, q^*] = \sum_i S_\alpha(X^{(i)})$; by homogeneity $S_\alpha(rY) = r^\alpha S_\alpha(Y)$ and by Lemma B.2 inside a cell, $S_\alpha(X^{(i)}) = \Theta(r_n^\alpha k_i^\theta)$ with $\theta = 1 - \alpha/d_\ell \in (0, 1)$ (the upper bound is deterministic; the lower bound holds for a positive fraction of cells by the law of large numbers over cells, which suffices for the sum). For balanced multiplicities $\sum_i k_i^\theta = \Theta(N \bar{k}^\theta) = \Theta(N^{1-\theta} n^\theta)$ (in general $n^\theta \leq \sum_i k_i^\theta \leq N^{1-\theta} n^\theta$ by subadditivity and Jensen). Taking logarithms: $\kappa_{\text{cell}} = -\alpha\gamma + \beta\alpha/d_\ell + 1 - \alpha/d_\ell$, and $\hat{d}_{\text{cell}} = \alpha / (\alpha\gamma + (1 - \beta)\alpha/d_\ell)$, in which α cancels.

(b) A cell with $k_i = 1$ contributes no edge; a cell with $2 \leq k_i \leq K$ contributes $k_i - 1$ edges of length $\Theta(r_n)$. Hence $S_\alpha[0, q^*] = c_n r_n^\alpha (n - N) = c_n r_n^\alpha (n - 1) q^*$ exactly, with c_n bounded above and below (this is where (a) breaks: its $\sum k_i^\theta$ counts 1 for a singleton cell whose true contribution is 0). Differentiating the logarithm with respect to $\log n$: $\kappa_{\text{cell}} = \varepsilon_n - \alpha\gamma + 1 + d \log q^* / d \log n + o(1)$. From $1 - q^* = (N - 1)/(n - 1)$, $d \log(1 - q^*) / d \log n = \beta_n - 1 + o(1)$, and $d \log q^* = -\frac{1-q^*}{q^*} d \log(1 - q^*)$, so $d \log q^* / d \log n = (1 - \beta_n)(1 - q^*) / q^*$. Substituting gives the formula; for $\gamma = 0$, $\varepsilon_n = 0$ we get $\kappa_{\text{cell}} - 1 = (1 - \beta_n)(1 - q^*) / q^* > 0$ and the stated $\hat{d}_{\text{cell}}$. Compatibility: $N \geq n/K$ gives $1 - q^* \geq (n/K - 1)/(n - 1) \rightarrow 1/K$, and $\log N \geq \log n - \log K$ gives an average $\beta_n \geq 1 - \log K / \log(n_2/n_1)$ on $[n_1, n_2]$. $\square$

Theorem B.5 (A mixed band does not interpolate). *Suppose a band contains edges of both strata, so that locally $S_\alpha[p_1, p_2](X_n) = An^{\kappa_1} + Bn^{\kappa_2}$ with $A, B > 0$, $\kappa_1 < \kappa_2$. Then (i) the local exponent $\kappa(n)$ lies in (κ_1, κ_2) ; (ii) if $\kappa_2 < 1$ then $\hat{d} \in (\hat{d}_1, \hat{d}_2)$; (iii) if $\kappa_1 < 1 < \kappa_2$ then $\hat{d} \in (\hat{d}_1, +\infty) \cup (-\infty, \hat{d}_2)$, i.e. outside the interval between the stratum estimates, including negative values and values below α .*

Proof. (i) $\kappa(n) = (\kappa_1 An^{\kappa_1} + \kappa_2 Bn^{\kappa_2}) / (An^{\kappa_1} + Bn^{\kappa_2})$ is a convex combination with positive weights. (ii) On $(\kappa_1, \kappa_2) \subset (-\infty, 1)$ the map $\kappa \mapsto \alpha / (1 - \kappa)$ is continuous and increasing. (iii) The map increases from $\hat{d}_1$ to $+\infty$ on $(\kappa_1, 1)$ and from $-\infty$ to $\hat{d}_2 < 0$ on $(1, \kappa_2)$. $\square$

Corollary B.6 (When text-like clouds are in case (iii)). If $\gamma = 0$, $\beta_n < 1$ and the sparse-regime premises hold, then $\kappa_{\text{fr}} < 1 < \kappa_{\text{cell}}$ by Theorems B.3 and B.4, so every fixed band that straddles q^* is in case (iii). When the band's composition itself drifts with n (a fixed band on a subsampling grid, since $q^*(n')$ moves), A and B depend on n and even (i) can fail: the fitted slope also measures the speed of the transition between strata.

Corollary B.7 (A fixed band switches stratum with q^*). The band $[q, 1]$ lies entirely in the frame stratum iff $q \geq q^*$; its relative position there is $u = (q - q^*) / (1 - q^*)$ with $\partial u / \partial q^* = (q - 1) / (1 - q^*)^2 < 0$. Under Hypothesis 2, $q^* = 1 - \text{TTR}$, so $[q, 1]$ is a frame band iff $\text{TTR} \geq 1 - q$, and $\partial u / \partial \text{TTR} = (1 - q) / \text{TTR}^2$ ($= 2$ at $q = \text{TTR} = 0.5$): the band moves by a fifth of the frame stratum per 0.1 of TTR at zero change in geometry.

Corollary B.8 (α -test). $\partial \hat{d}_{\text{fr}} / \partial \alpha = d_g^2(1 - \beta) / (d_g(1 - \beta) + \alpha\beta)^2$, which vanishes iff $\beta = 1$. A true dimension does not depend on α ; a drift of $\hat{d}$ with $\alpha \in \{0.5, 1, 2\}$ beyond the finite-sample baseline (measured on a homogeneous cloud of the same effective size N' : $\times 1.4$ at $N' \approx 200$, $\times 1.1$ at $N' \approx 200\text{--}470$, $d \approx 24$) signals $\beta < 1$ or a mixed band. The test is one-sided: mixtures also produce α -drift.

Corollary B.9 (Sign test). For a d -regular stratum with $d > \alpha$, $\kappa = 1 - \alpha/d \in (0, 1)$ and $\hat{d} = d > \alpha$. Hence $\hat{d}_{[p_1, p_2]} \leq \alpha \iff \kappa_{[p_1, p_2]} \notin (0, 1]$, and such a band estimates no dimension $d > \alpha$; in particular $\hat{d} < 0$ means super-linear energy growth, impossible for a homogeneous stratum. Caveat: a genuine stratum with $d \leq \alpha$ is also rejected; repeat at smaller α to tell the two apart.

Proposition B.10 (Corrected estimator). (i) Place bands as $[v_1 q^*, v_2 q^*]$ and $[q^* + u_1(1 - q^*), q^* + u_2(1 - q^*)]$ with q^* recomputed on each subsample; by Theorem 3.1(v) each band is then inside one stratum at every n' , removing the pathologies of Theorem B.5. (ii) Fit the frame band against $\log N'$: with β the slope of $\log N'$ on $\log n'$ over the same grid (exactly linear), $\kappa_{\text{fr}}^{(N)} = \kappa_{\text{fr}} / \beta = 1 - \alpha/d_g$, so $\alpha / (1 - \kappa_{\text{fr}}^{(N)}) = d_g$; this coincides with solving Theorem B.3 for d_g , $d_g = \alpha \beta \hat{d}_{\text{fr}} / (\alpha - \hat{d}_{\text{fr}}(1 - \beta))$. Isolation alone (i) is not enough: it turns the confound in q^* into a confound in β .

Corollary B.11 (Conditioning). In units of the fitted slope the correction adds no amplification: for both classical PHD and the corrected estimator $\partial \log d / \partial \log(\text{slope}) = d/\alpha - 1$ ($= 23$ at $d = 24$, $\alpha = 1$). In units of the uncorrected estimate the amplification is $1 + \frac{d_g}{\alpha} \frac{1 - \beta}{\beta}$, which exceeds $d/\alpha - 1$ once $\beta < \beta_0 = 1 / (2(1 - \alpha/d_g))$ (0.52 at $d_g = 24$). Consequence: a systematic 1% drift of the frame energy along the grid shifts $d_g \approx 24$ by about a third.

Remark B.12 (Class A and class B experiments). All the confounds above sit in n , N , q^* , β . If two clouds are built from the same text (layers, checkpoints, quantisation), these coincide exactly and any difference in $\hat{d}$ is geometric — although the uncorrected frame estimate still responds to d_g with elasticity ≈ 0.1 , so the correction is needed for power. If the texts differ (human vs AI, domain vs domain), the counting quantities differ systematically, and a signal has to be checked against a lexical baseline or on lexically matched pairs, as in Section 5.

C Synthetic checks in brief

Cell clouds were built exactly as in Definition 2.5 (frame: uniform in a d_g -ball embedded in $\mathbb{R}^{100}$; cells: uniform in d_ℓ -balls of radius r), the MST computed exactly, and exponents fitted over a geometric grid of subsample sizes, averaging κ (not $\hat{d}$) over seeds. Full tables and scripts are in the theory report and its code/ directory; the numbers that matter for this note are these.

- **Exact split.** With $n = 1030$, $N = 600$ ($q^* = 0.418$) the share of within-cell edges is exactly 1.000 in every 5% bin below q^* , exactly 0.000 above, and 0.365 in the bin $[0.40, 0.45]$ — against the predicted $(0.418 - 0.40) / 0.05 = 0.36$.
- **Frame band formula.** Over eight configurations (β from 0.01 to 0.78, $d_g \in \{12, 24\}$) measured and predicted $\kappa_{\text{fr}} = \beta(1 - \alpha/d_g)$ differ by at most 0.014; with large cells ($\beta \rightarrow 0$) the frame band returns $\hat{d} \rightarrow 1$ although $d_g = 24$, exactly as Theorem B.3 says.

- **Cell band, sparse regime.** $\kappa_{\text{cell}} > 1$ (negative $\hat{d}$) in every grid node of every sparse configuration; the local formula of Theorem B.4(b) matches node by node within 0.07–0.08 in κ , the residual having the sign predicted by the composition drift ε_n .
- **Counting confound at frozen geometry** ($n = 201$, $d_g = 24$, N from 50 to 170). Classical PHD ranges $\times 4.8$, the fixed band $[0.5, 1] \times 8.1$ (it jumps from 5.5 to 14.3 exactly where q^* crosses 0.5), the relative frame band fitted on $\log n' \times 5.4$, and the same band fitted on $\log N' \times 1.32$ with values 16–22 near the ceiling (≈ 21) of what 110–170 points in 24 dimensions can give. The relative cell band is negative in every row.
- **Geometric signal at fixed counts.** Elasticity $\partial \log \hat{d} / \partial \log d_g$: classical PHD 0.06, fixed $[0.5, 1]$ 0.66, relative frame band on $\log n'$ 0.10, on $\log N'$ 0.89 (ideal 1).
- **Residual of the corrected estimator.** Its remaining $\times 1.32$ response to N is entirely the single-linkage perturbation of Proposition 3.2: the MST on the true centres gives 20–21.5 for all $N \geq 80$; the frame energy differs from it by only 0.2–2.6%, but that difference drifts along the grid, and at $d \approx 24$ a 1% drift is a third of the estimate (Corollary B.11). Building the frame on the empirical cell centroids instead of single linkage halves the residual ($\times 1.14$).

References

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